

Enterprise Supply Chain Testing Platform

Project Overview

Project Title

Enterprise Supply Chain Testing Platform

Objective

The objective of this project is to build an AI-powered multi-agent platform that automates supply chain resilience testing by simulating real-world disruption scenarios, analyzing operational risks, validating business data, verifying regulatory compliance, and generating executive reports. The platform enables organizations to proactively identify vulnerabilities and improve decision-making before actual supply chain disruptions occur.

Business Problem

Modern supply chains are highly dependent on suppliers, transportation networks, warehouses, and inventory systems. Unexpected disruptions such as supplier failures, transportation delays, inventory shortages, cyberattacks, and natural disasters can significantly impact business operations. Traditional supply chain testing is often manual, time-consuming, and unable to provide timely insights. This project aims to automate the entire supply chain testing process using AI agents, enabling organizations to assess risks, improve resilience, and ensure business continuity.

Expected Outcomes

- Automated supply chain resilience testing
- AI-driven test planning and scenario generation
- Operational risk identification and assessment
- Automated data validation and quality checks
- Compliance verification against industry standards and internal policies
- Intelligent simulation of supply chain disruptions
- Executive-level reporting with actionable recommendations
- Improved business continuity and decision-making

2. Dataset Details

Dataset Name

Synthetic Supply Chain Knowledge Base

Source

The datasets used in this project are **synthetically generated** to simulate real-world supply chain operations. They were created specifically for testing and evaluating supply chain resilience under different disruption scenarios. The synthetic data closely resembles enterprise supply chain data while avoiding the use of confidential or proprietary business information.

Dataset Overview

The knowledge base consists of multiple datasets representing different components of a supply chain ecosystem, including suppliers, inventory, customer orders, shipments, and disruption events. These datasets are used by the AI agents to retrieve relevant business information, simulate disruption scenarios, perform risk analysis, validate data quality, verify compliance, and generate executive reports.

The datasets are stored as CSV files and converted into vector embeddings using **Sentence Transformers** before being stored in **ChromaDB**, enabling semantic retrieval through a Retrieval-Augmented Generation (RAG) pipeline.

Datasets Used

Dataset	Description
inventory.csv	Stores product inventory details such as SKU, warehouse location, available stock, and reorder point.
suppliers.csv	Contains supplier information including supplier ID, country, risk score, and supplier details.
orders.csv	Stores customer order information used for demand and fulfillment analysis.
shipments.csv	Contains shipment details such as shipment ID, carrier, delivery status, and transportation information.
disruptions.csv	Includes disruption events such as supplier failures, port strikes, cyberattacks, natural disasters, and transportation delays used for simulation testing.

Purpose of the Dataset

The synthetic datasets provide realistic business scenarios that enable the platform to:

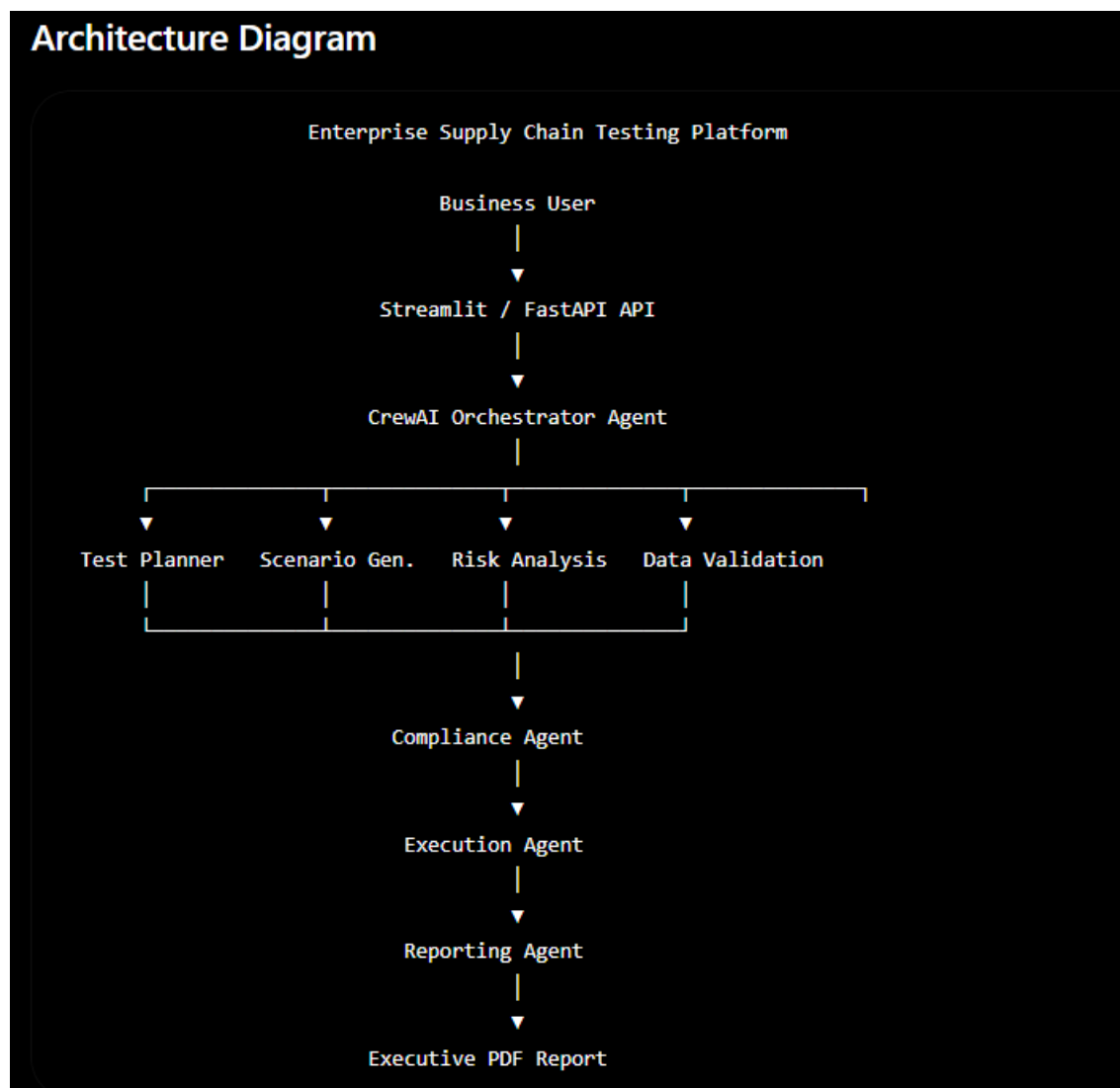
- Simulate real-world supply chain disruptions.

- Retrieve contextual information through the RAG pipeline.
- Perform risk assessment and operational analysis.
- Validate data quality before execution.
- Support compliance verification.
- Generate intelligent recommendations and executive reports.

System Architecture

The **Enterprise Supply Chain Testing Platform** is built using a **Multi-Agent AI Architecture** that automates supply chain resilience testing. The platform uses **CrewAI** to orchestrate multiple AI agents, **PydanticAI** for structured outputs, **Groq/HuggingFace LLMs** for intelligent reasoning, **ChromaDB** as the knowledge base, and **FastAPI** for API communication. Each agent performs a specific task, including test planning, scenario generation, risk analysis, data validation, compliance verification, execution, and report generation.

Architecture Diagram



Key Components

- **FastAPI** – Receives user requests and exposes REST APIs.
- **CrewAI** – Coordinates the workflow between AI agents.
- **Specialized AI Agents** – Perform planning, risk analysis, validation, compliance, execution, and reporting.
- **ChromaDB** – Stores and retrieves supply chain knowledge.
- **Groq/HuggingFace LLM** – Provides intelligent reasoning and recommendations.
- **Reporting Module** – Generates executive reports in PDF format.

Multi-Agent Architecture

The Enterprise Supply Chain Testing Platform follows a **multi-agent architecture**, where each AI agent is responsible for a specific task in the supply chain testing workflow. This modular design improves scalability, maintainability, and decision-making by allowing each agent to focus on a specialized function.

1. Orchestrator Agent

The **Orchestrator Agent** acts as the central controller of the platform. It receives the testing request from the user, analyzes the business objective, and coordinates the complete workflow by assigning tasks to the appropriate AI agents. It monitors the execution process, manages communication between agents, and consolidates the final results into a structured workflow.

Responsibilities

- Receive testing requests.
- Coordinate all AI agents.
- Manage workflow execution.
- Collect and combine outputs.

Test Planner Agent

The **Test Planner Agent** converts the business objective into a structured testing strategy. It defines the testing objective, selects appropriate disruption scenarios, and establishes the success criteria required to evaluate supply chain resilience.

Responsibilities

- Create test plans.
- Define testing objectives.

- Select disruption scenarios.
- Generate success criteria.

Scenario Generator Agent

The **Scenario Generator Agent** creates realistic supply chain disruption scenarios using the knowledge base and LLM. It simulates events such as supplier failures, transportation delays, inventory shortages, cyberattacks, and natural disasters to evaluate how the supply chain performs under different conditions.

Responsibilities

- Generate disruption scenarios.
- Simulate supplier failures.
- Create logistics disruptions.
- Produce edge-case scenarios.

Risk Analysis Agent

The **Risk Analysis Agent** evaluates the operational and business impact of the generated scenarios. It analyzes supplier risks, inventory shortages, transportation issues, financial impact, and operational vulnerabilities. Based on the assessment, it recommends mitigation strategies to reduce business risk.

Responsibilities

- Assess operational risks.
- Identify vulnerable areas.
- Calculate risk severity.
- Recommend mitigation strategies.

Data Validation Agent

The **Data Validation Agent** verifies the quality and consistency of the supply chain datasets before executing the simulation. It checks for missing values, duplicate records, invalid lead times, and business rule violations to ensure accurate testing results.

Responsibilities

- Validate datasets.
- Detect missing values.
- Remove duplicate records.

- Verify business rules.

Compliance Agent

The **Compliance Agent** ensures that the testing process and generated recommendations comply with industry standards and organizational policies. It validates predefined compliance rules such as **ISO 9001**, **ISO 28000**, **SOX**, and internal supply chain policies before approving the final results.

Responsibilities

- Verify compliance rules.
- Validate regulatory standards.
- Check internal policies.
- Generate compliance summary.

Execution Agent

The **Execution Agent** performs the actual supply chain simulation based on the generated scenarios. It executes the test plan, simulates disruptions, analyzes operational outcomes, and records the execution results for reporting.

Responsibilities

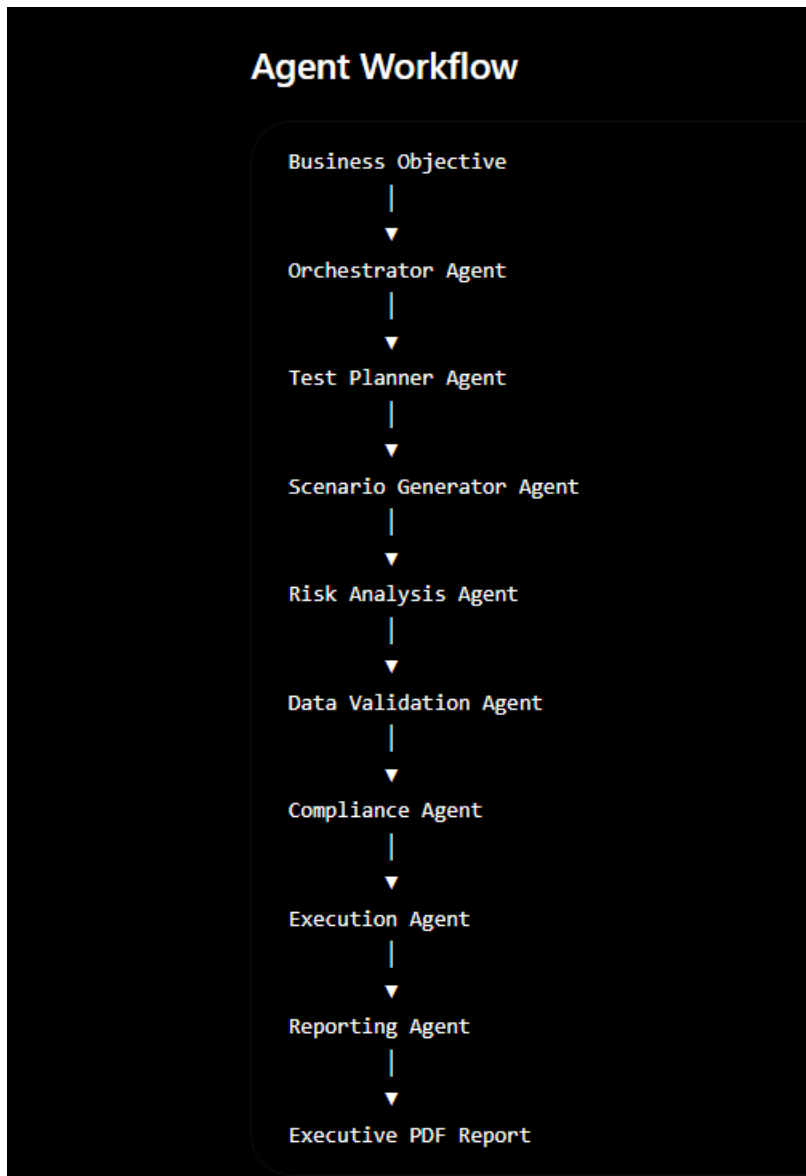
- Execute test scenarios.
- Simulate disruptions.
- Capture execution results.
- Monitor simulation status.

Reporting Agent

The **Reporting Agent** compiles the outputs generated by all previous agents into a comprehensive executive report. The report includes the testing objective, risk assessment, validation results, compliance findings, execution summary, remediation plan, and business recommendations for decision-makers.

Responsibilities

- Generate executive reports.
- Summarize risk findings.
- Present compliance results.
- Recommend mitigation actions.



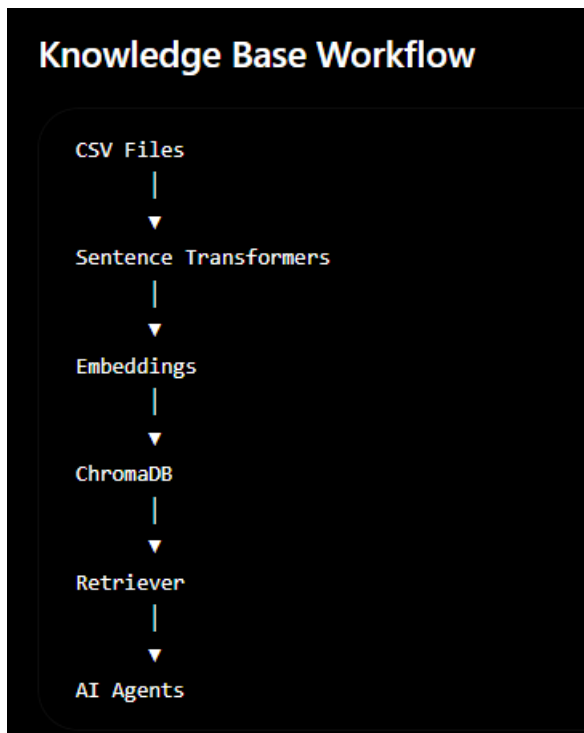
Knowledge Base

The **Knowledge Base** provides the contextual information required by the AI agents to perform supply chain resilience testing. It consists of synthetic datasets representing different supply chain entities, including suppliers, inventory, orders, shipments, and disruption events. These datasets simulate real-world business scenarios while avoiding the use of confidential enterprise data.

The data is converted into vector embeddings using **Sentence Transformers** and stored in **ChromaDB**. When an AI agent requires information, the system retrieves the most relevant context from the vector database through a **Retrieval-Augmented Generation (RAG)** pipeline. This enables the LLM to generate accurate and context-aware responses instead of relying only on its pre-trained knowledge.

Knowledge Base Datasets

Dataset	Description
inventory.csv	Stores product inventory details, warehouse locations, stock levels, and reorder points.
suppliers.csv	Contains supplier information, country, and risk scores.
orders.csv	Stores customer order details for demand analysis.
shipments.csv	Contains shipment status, carrier information, and delivery details.
disruptions.csv	Includes disruption events such as supplier failures, port strikes, cyberattacks, and natural disasters.



Challenges

During the development of the Enterprise Supply Chain Testing Platform, several technical and architectural challenges were encountered.

Challenges Faced

- Designing and coordinating a multi-agent workflow using CrewAI.
- Generating consistent and structured AI responses using Large Language Models.
- Building a Retrieval-Augmented Generation (RAG) pipeline with ChromaDB.
- Creating realistic synthetic supply chain datasets for testing.

- Ensuring data quality through validation before simulation.
- Implementing compliance verification using predefined business rules.
- Generating clear and business-friendly executive reports from multiple agent outputs.

Learnings

This project provided practical experience in developing enterprise AI applications using modern frameworks and multi-agent architectures.

Key Learnings

- Learned to design and implement Multi-Agent AI systems using CrewAI.
- Gained hands-on experience with PydanticAI for structured AI outputs.
- Built a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and Sentence Transformers.
- Improved understanding of supply chain resilience testing and risk assessment.
- Learned to integrate FastAPI with AI agents for REST-based applications.
- Developed skills in executive report generation and enterprise workflow automation.

Conclusion

The **Enterprise Supply Chain Testing Platform** successfully demonstrates how Artificial Intelligence can automate and improve supply chain resilience testing. By integrating **CrewAI** for agent orchestration, **PydanticAI** for structured outputs, **Groq/HuggingFace LLMs** for intelligent reasoning, and **ChromaDB** for knowledge retrieval, the platform provides a scalable and efficient solution for supply chain risk assessment.

The system enables organizations to simulate real-world disruptions, identify operational risks, validate data quality, verify compliance, and generate executive-level reports with actionable recommendations. This project highlights the potential of AI-driven multi-agent systems to support proactive decision-making, improve business continuity, and strengthen overall supply chain resilience.